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Assessing National Trends and Perceived Safety of Off-Label Ciprofloxacin-Dexamethasone Use by Pediatric Otolaryngologists

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Objectives: Off-label use of Ciprodex® (ciprofloxacin-dexamethasone: CPD), an antibiotic-steroid combination solution, in the airway has been reported by pediatric otolaryngologists with anecdotal success. We examined national trends regarding off-label CPD use including prevalence, common indications, prescriber patterns, adverse events, and accessibility.

Methods: 15-item cross-sectional survey was distributed to American Society of Pediatric Otolaryngology members from January–April 2022. Univariate analyses were performed to compare responses for users of off-label CPD versus non-users. Ease of access was compared across geographies and practice types using multivariate logistic regressions.

Results: Of the 163 complete responses (26.6% response rate), 156 (95.7%) reported using off-label CPD. Most common indications for off-label CPD were tracheal granulation (87.8%, $n = 137$) and choanal atresia (82.1%, $n = 128$). Ease of access was significantly increased in the Midwest (OR:18.79, 95%CI:3.63–1.24, $p = 0.001$) and West (OR:29.92, 95%CI:3.55–682.00, $p = 0.006$). Ease of access was significantly lower at tertiary referral centers (OR:0.11, 95%CI:0.01–0.64, $p = 0.041$) and private practices (OR:0.04, 95%CI:0.002–0.33, $p = 0.009$) compared to academic free-standing children's hospitals. Two-thirds of respondents reported feeling "Very Comfortable" with the safety profile of off-label CPD; 99.4% ($n = 156$) felt that the benefits outweighed the risks of off-label use. Seven respondents (4.5%) reported adverse events (e.g., local allergic reaction, cushingoid symptoms) from off-label use.

Conclusions: Our findings (26.6% response rate) suggest that off-label CPD is commonly used by pediatric otolaryngologists, many of whom reported feeling that the benefits of off-label CPD outweigh the risks. Our results establish a baseline for future efforts to assess the efficacy and safety of off-label CPD and to improve its accessibility.

Key Words: aerodigestive, airway surgery, Ciprodex, off-label, pediatric, tracheostomy.

Level of Evidence: V

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INTRODUCTION

The term "off-label" describes any drug use not included in the U.S. Food and Drug Administration (FDA)-approved labeling of a medication.[1] Physicians prescribe medications off-label for numerous reasons, such as patient age, desired dosage higher or lower than the FDA-designated dosage, or indications for which the drug is prescribed.[2] Provider experience and patient preference are other common reasons for off-label prescribing. Because patients <18 years are often excluded from drug trials, there is frequent off-label drug use in pediatrics. The proportion of pediatric patients exposed to at least one off-label medication in outpatient and inpatient settings has been reported to be as high as 40% and 97% respectively.[2]

In pediatric otolaryngology, Ciprodex® (ciprofloxacin-dexamethasone: CPD) is frequently prescribed off-label despite limited research regarding its efficacy. CPD drops are a topical fluoroquinolone antibiotic and corticosteroid combination solution effective in controlling granulation tissue and decreasing edema.[3,4] CPD is FDA-approved for the management of (1) acute otitis media (AOM) with rupture or with tympanostomy tubes and (2) acute otitis externa in patients ≥6 months.[5] CPD has been found to be effective for treating granulation in children with AOM with otorrhea through tympanostomy tubes and superior to ofloxacin for this indication.[6] In vitro studies have shown that ciprofloxacin increases ciliary beat

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Additional supporting information may be found in the online version of this article.

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Dr. Rutter has the following disclosures: (1) He has a suprastomal stent named after him, marketed by Boston Medical Products, for which he has declined royalties; (2) He is the patent holder for the Aeris airway balloon marketed by Bryan Medical, and is a consultant for Bryan Medical. He receives licensing fees for this product; (3) He is a consultant for Tracoe (now ATOS), a tracheotomy tube company.

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frequency in a dose-dependent manner, likely by facilitating Calcium influx through voltage-dependent channels.[7] CPD is used for conditions involving mucosal scarring and granulation, such as nasal surgery, tracheostomy management, airway stent removal, or endoscopic airway surgeries that breach the mucosa.[4] Several routes of administration are available for CPD: topical drops, drops to a stoma, drops through endotracheal or tracheostomy tubes, and nebulized inhaled solution.[6,8–10]

There are little data regarding the use or efficacy of off-label CPD. Many prescribers feel CPD is so effective that withholding the medication for randomized trials would be unethical. Given the anecdotal safety and efficacy of CPD, we aimed to collect objective data regarding prescriber patterns, off-label indications, and accessibility of off-label CPD to establish a baseline for future research. We hope that our results will promote awareness of the utility of CPD for off-label indications and provide data to support its addition to hospital formularies for off-label indications and routes of administration.

MATERIALS AND METHODS

Survey Development and Distribution

A 15-item, 3-minute survey (**Supplement**) to assess pediatric otolaryngologists' utilization of and experience with off-label CPD was developed using Qualtrics (Provo, UT). The survey was reviewed, approved, and distributed by the American Society of Pediatric Otolaryngology (ASPO) to its members on two occasions, 2 months apart, from January–April 2022. The survey was completed voluntarily and anonymously; no incentives were distributed. Yale Institutional Review Board deemed the study exempt.

Survey Description

Initial four questions established by ASPO included number of years in practice, fellowship status, practice setting, and geographic location. Subsequent questions addressed utilization of off-label CPD, which was defined for respondents as "nebulized or topical CPD for indications besides otitis externa and tube otorrhea." Our survey questions did not specify or differentiate between brand name and generic CPD. Survey questions covered frequency of use, routes of administration, indications, prescriber comfort with safety profile, experience with adverse reactions, and ease of access in practice setting. Non-users were asked to indicate their reasons for not using the medication.

Statistical Analyses

Statistical analyses were performed using Rstudio v4.1.0 (Boston, MA) and GraphPad Prism v9.2.0 (San Diego, CA). Geographic regions in the United States (U.S.) were defined as West, Midwest, Northeast, South, or Southwest using U.S. Census Bureau divisions,[11] and non-U.S. locations were grouped as 'Other'. Chi-squared and Fisher's exact tests were performed to compare categorical variables for off-label CPD users versus non-users and based on years in practice. Chi-squared tests were performed to compare ease of access to off-label CPD across geographic regions and practice types. Comfort level and ease of access responses were categorized into binary outcomes. For example, ease of access Likert scores from 1 ("not easy at all") to 4 ("very easy") were divided into two categories: "not easy to access" (1, 2) and "easy to access" (3, 4). Multivariate logistic regressions were used to elucidate factors impacting comfort level and ease of access. For open-ended questions, responses were analyzed qualitatively. Common themes were identified and clarifications of responses to closed-ended questions were noted.

RESULTS

One-hundred and sixty-three respondents completed the survey, yielding a response rate of 26.6% (Table I). Of these, 156 (95.7%) indicated off-label CPD use with 47.4% ($n = 74$) reporting that they have been prescribing off-label for >10 years. Seven respondents indicated that they do not use off-label CPD. All seven respondents had been practicing for >10 years, and three practiced outside the United States. Of these, four stated that there is not enough data to support its use, one had not heard of off-label indications, and two indicated other reasons—difficulty accessing it and preference for use only after obtaining culture and sensitivity data.

Indications and Route of Administration for Off-Label CPD

A majority (80.8%) of respondents reported using off-label CPD 1–5 times per month. Indications for off-label CPD use differed significantly based on years in practice (Table II). Percentage of respondents who use CPD for nasal congestion was significantly higher amongst those practicing for <5 years and 5–10 years compared to those practicing for >10 years (54.8% vs. 46.4% vs. 27.8%, $p = 0.012$). Percentage of respondents using CPD for stomal site granulation and tracheitis was significantly greater for those in practice <5 years and 5–10 years relative to >10 years in practice (Table II). Significantly greater proportions of respondents practicing for <5 years and 5–10 years reported using nebulized CPD compared to those practicing for >10 years (87.1% vs. 71.4% vs. 49.5%, respectively, $p < 0.001$); similar differences were observed for topical drops to tracheostomy stoma (96.8% vs. 75.0% vs. 61.9%, $p < 0.001$).

Ease of Access to Off-Label CPD

On univariate analysis, ease of access to off-label CPD was significantly different across geographic locations with 78.1% ($n = 32$) of respondents practicing in the Midwest reporting 'Very easy' access compared to 43.8% ($n = 14$) of respondents in the Northeast ($p = 0.003$) (Fig. 1). On multivariate analysis (MVA), likelihood of easy access was significantly increased in the Midwest (Odds ratio (OR):18.79, 95% confidence interval (CI):3.63–1.24, $p = 0.001$), the South (OR:5.65, 95% CI:1.14–32.10, $p = 0.040$), and the West (OR:29.92, 95% CI:3.55–682.00, $p = 0.006$) (Fig 2, Supplementary Table 1).

A greater percentage (70.3%) of respondents practicing at academic freestanding children's hospitals reported "Very easy" access to off-label CPD compared to those at academic tertiary referral centers and in private practice (51.8% vs. 42.1%, respectively). However, this difference

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TABLE I.
Comparison of Demographic and Practice Characteristics for Pediatric Otolaryngologists Who Do and Do Not Use Off-Label Ciprofloxacin-Dexamethasone.

	No Off-Label Use $N = 7$	Yes Off-Label Use $N = 156$	p-Value
Fellowship completed, yes (%)	6 (85.7)	154 (98.7)	0.286
Type of practice (n,%)			0.947
Academic Practice—Freestanding Children's Hospital	1 (14.3)	37 (23.7)	
Academic Practice—Tertiary Referral Center	5 (71.4)	83 (53.2)	
Private Practice—Freestanding Children's Hospital	1 (14.3)	16 (10.3)	
Private Practice—General Hospital	0 (0.0)	3 (1.9)	
Hospital Employed	0 (0.0)	14 (9.0)	
Non-University Teaching Hospital	0 (0.0)	1 (0.6)	
Other	0 (0.0)	2 (1.3)	
Number of years in practice (n,%)			0.126
<5 years	0 (0.0)	31 (19.9)	
5–10 years	0 (0.0)	28 (17.9)	
>10 years	7 (100.0)	97 (62.2)	
Number of years off-label use (n,%)			-
<5 years	—	35 (22.4)	
5–10 years	—	47 (30.1)	
>10 years	—	74 (47.4)	
Geographic location			**0.004**
Northeast	2 (28.6)	32 (20.5)	
Midwest	0 (0.0)	41 (26.3)	
South	2 (28.6)	24 (15.4)	
Southwest	0 (0.0)	26 (16.7)	
West	0 (0.0)	21 (13.5)	
Canada	0 (0.0)	4 (2.5)	
Other	3 (42.9)	8 (5.1)	

Note: Bold p-Values = statistically significant ($p < 0.05$).

did not reach statistical significance ($p = 0.059$) (Fig. 3). On MVA, likelihood of easy access was significantly decreased at tertiary referral centers (OR:0.11, 95% CI:0.01–0.64, $p = 0.041$) and in private practice (OR:0.04, 95%CI:0.002–0.33, $p = 0.009$) (Fig. 2).

Safety Profile of Off-Label CPD

Of the 156 pediatric otolaryngologists who reported off-label CPD use, two-thirds ($n = 104$) reported feeling "Very Comfortable" with the safety profile of off-label CPD (Figure 4). Although 14.7% ($n = 23$) reported feeling "Very Uncomfortable" with the safety profile of CPD, no adverse patient events were reported by this group, all of them use the medication 1–5 times per month, and all felt that the benefits outweigh the risks. Overall, 99.4% of pediatric otolaryngologists felt that the benefits of off-label usage outweigh the risks. No respondent characteristics were significantly associated with comfort level on MVA (Suppl. Table 2).

Seven respondents (4.5%), all of whom were practicing for >10 years, reported adverse events (AEs) from off-label use. Four respondents had patients with Cushingoid symptoms (e.g., moon facies), one had a patient with a local reaction/possible allergy, one had a patient with steroid-related mental status changes, and one had a neonatal patient with adrenal suppression after nasal application. A majority of respondents (85.7%, $n = 6$) reported that the frequency of AEs was once every few years (Table III). 71.4% reported that the severity of these AEs was "Not severe at all" or "mildly severe"; none reported a "very severe" event.

DISCUSSION

This study examined national trends regarding the use of off-label CPD by pediatric otolaryngologists including prevalence, common indications, route of administration, prescriber patterns, adverse events, and accessibility. Of the 163 survey respondents, 95.7% use CPD off-label and 99.4% feel that the benefits outweigh the risks. There were only seven non-users (4.3%), four of whom practiced in the United States. The main reason cited by non-users was lack of data. While 76% of pediatric otolaryngologists indicated easy access to off-label CPD, the remainder said it was "Not easy at all" (13%) or "Somewhat difficult" (11%) to access at their institution or practice.

Our findings indicate that off-label CPD use is common across the United States and felt to be safe by many

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TABLE II.
Overview of Demographics and Off-Label Ciprofloxacin-Dexamethasone Utilization Patterns Comparing Respondents By Number of Years in Practice.

$N = 156$	All $N = 156$	<5 years $N = 31$	5–10 years $N = 28$	>10 years $N = 97$	p-Value
Fellowship completed, yes (%)	154 (98.7)	31 (100.0)	28 (100.0)	95 (97.9)	0.540
Type of practice					0.126
Academic Practice—Freestanding Children's Hospital	37 (23.7)	6 (19.4)	7 (25.0)	24 (24.7)	
Academic Practice—Tertiary Referral Center	83 (53.2)	23 (74.2)	17 (60.7)	43 (44.3)	
Private Practice—Freestanding Children's Hospital	16 (10.2)	2 (6.5)	0 (0.0)	14 (14.4)	
Private Practice—General Hospital	3 (1.9)	0 (0.0)	1 (3.6)	2 (2.1)	
Hospital Employed	14 (9.0)	0 (0.0)	3 (10.7)	11 (11.3)	
Non-University Teaching Hospital	1 (0.7)	0 (0.0)	0 (0.0)	1 (1.0)	
Other	2 (1.3)	0 (0.0)	0 (0.0)	2 (2.1)	
Geographic location					**0.012**
Northeast	32 (20.5)	8 (25.8)	6 (21.4)	18 (18.6)	
Midwest	41 (26.2)	10 (32.3)	7 (25.0)	24 (24.7)	
South	24 (15.4)	4 (12.9)	4 (14.3)	16 (16.5)	
Southwest	26 (16.7)	3 (9.7)	0 (0.0)	23 (23.7)	
West	21 (13.5)	3 (9.7)	10 (35.7)	8 (8.2)	
Canada	4 (2.6)	2 (6.5)	0 (0.0)	2 (2.1)	
Other	8 (5.1)	1 (3.2)	1 (3.6)	6 (6.2)	
Frequency of off-label use					0.168
>10 times/month	10 (6.4)	2 (6.5)	3 (10.7)	5 (5.2)	
1–5 times/month	126 (80.8)	25 (80.6)	18 (64.3)	83 (85.6)	
5–10 times/month	20 (12.8)	4 (12.9)	7 (25.0)	9 (9.3)	
Setting of off-label use					0.657
Mean % of time inpatient (SD)	49.6 (29.7)	52.4 (27.4)	45.4 (26.3)	49.9 (31.4)	
Mean % of time outpatient (SD)	50.4 (29.7)	47.6 (27.4)	54.6 (26.3)	50.1 (31.4)	
Route of administration*					
Nebulized	95 (60.9)	27 (87.1)	20 (71.4)	48 (49.5)	**<0.001**
Topical drops through a tracheostomy tube	130 (83.3)	29 (93.5)	22 (78.6)	79 (81.4)	0.219
Topical drops to a tracheostomy stoma	111 (71.2)	30 (96.8)	21 (75.0)	60 (61.9)	**<0.001**
Topical nasal drops	134 (85.9)	28 (90.3)	26 (92.9)	80 (82.5)	0.278
Other	10 (6.4)	2 (6.5)	3 (10.7)	5 (5.2)	0.571
Indications*					
Choanal atresia	128 (82.1)	24 (77.4)	25 (89.3)	79 (81.4)	0.479
Following airway surgery	105 (67.3)	24 (77.4)	21 (75.0)	60 (61.9)	0.174
Nasal congestion	57 (36.5)	17 (54.8)	13 (46.4)	27 (27.8)	**0.012**
Stomal site granulation	108 (69.2)	30 (96.8)	21 (75.0)	57 (58.8)	**<0.001**
Subglottic stenosis	63 (40.4)	16 (51.6)	12 (42.9)	35 (36.1)	0.295
Tracheal granulation	137 (87.8)	30 (96.8)	25 (89.3)	82 (84.5)	0.187
Tracheitis	106 (67.9)	27 (87.1)	21 (75.0)	58 (59.8)	**0.012**
Other	18 (11.5)	5 (16.1)	1 (3.6)	12 (12.4)	0.294

*$N = 480$ for route of administration and $N = 731$ for indications because respondents were able to *select all that apply*.
Note: Bold p-Values = statistically significant ($p < 0.05$).
Abbreviation: SD = standard deviation.

of the surveyed pediatric otolaryngologists. However, there remains a significant proportion of pediatric otolaryngologists practicing in select geographic regions, tertiary referrals centers, and private practices who do not have easy access to CPD for off-label indications. To our knowledge, this study is the first to examine access to off-label CPD and patterns of utilization nationally. We hope our findings can guide the efforts of interdisciplinary teams of health care providers, pharmacists, and hospital administrators to facilitate easier access to CPD for off-label use across institutions.

While there is limited literature on off-label CPD use, the few studies that have investigated its utility are consistent with the findings of our survey. Much of the literature surrounding off-label CPD use is in the setting of airway surgery including reconstruction, tracheal trauma

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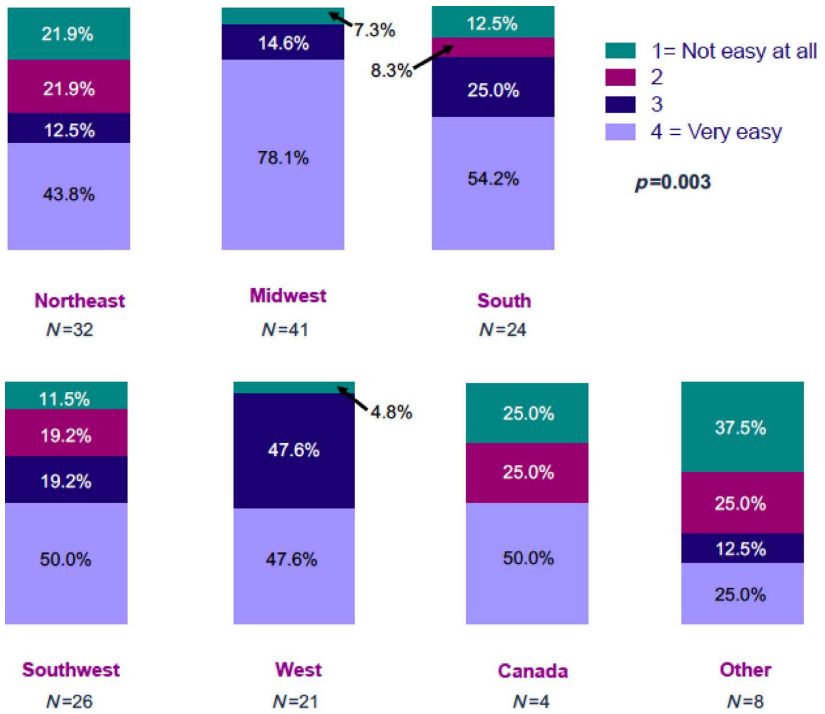

Fig. 1. Ease of access to off-label ciprofloxacin-dexamethasone by geographic region. [Color figure can be viewed in the online issue, which is available at www.laryngoscope.com.]

management, endoscopic procedures, and bronchoscopic procedures such as foreign body removal with proximal bronchoplasty.[4,8,12–14] A 25-year retrospective review study at Lurie Children's Hospital in Chicago, IL compared outcomes for patients who underwent slide tracheoplasty or tracheal resection before ($n = 20$) and after ($n = 21$) the initiation of routine post-operative administration of off-label nebulized CPD in 2007.[8] In this study, Stephens et al. found a decrease in median length of stay from 30 to 18 days once nebulized CPD was introduced. They also found that the number of post-operative bronchoscopy procedures required to address granulation tissue decreased from a range of 1–4 prior to the initiation of CPD compared to 0 after its introduction.[8] These findings are also reflected in our survey data: tracheal granulation was the most common off-label indication reported by 87.8% of pediatric otolaryngologists. The second most common indication was choanal atresia, reported by 82.1% of respondents. Ben-Dov et al. reported the use of CPD off-label to prevent postoperative granulation after choanal atresia repair in an infant.[15] One potential explanation for the limited data on the use of CPD or other nebulized antibiotic-steroid combinations may be that because these medications are so effective that clinicians may not feel comfortable omitting their use solely for research purposes. Given this, our survey results provide insights into the numerous indications for which pediatric otolaryngologists are commonly using CPD off-label.

In addition to widespread use for various off-label indications, pediatric otolaryngologists reported utilizing different routes of administration for CPD. Although most respondents use topical drops for various indications, 60.9% also use nebulized CPD. For children with tracheostomies, Rutter et al. note their dosing at 3 drops thrice daily for 5–7 days through the tracheostomy tube.[4] Studies have reported dosing nebulized CPD at 1 mL (20 drops) and 0.5 mL (10 drops) in 1 mL normal saline 2–3 times daily for 7 days after tracheal surgery for patients ≥2 years-old and <2 years-old, respectively.[4,8,12,16] Each mL of CPD contains 3 mg ciprofloxacin and 1 mg dexamethasone; one drop contains 0.04 mg ciprofloxacin and 0.11 mg dexamethasone.[5] Anecdotal success of nebulized CPD has been reported for airway foreign body retrieval[13], laryngeal trauma management[14], tracheoplasty and tracheal resection[8], and endoscopic airway management in children.[4,16] Notably, in our survey, a significantly greater percentage of respondents practicing for <10 years use nebulized CPD compared to those practicing for >10 years. This trend may be explained by greater exposure to off-label CPD use for pediatric otolaryngologists who completed training more recently. Our survey findings, in the context of anecdotal success reported by several studies across institutions, highlight the need for efforts to generate quantitative data on the safety and efficacy of off-label topical and nebulized CPD use.

Ease of access, which can impact off-label utilization, was found to be significantly different across geographic

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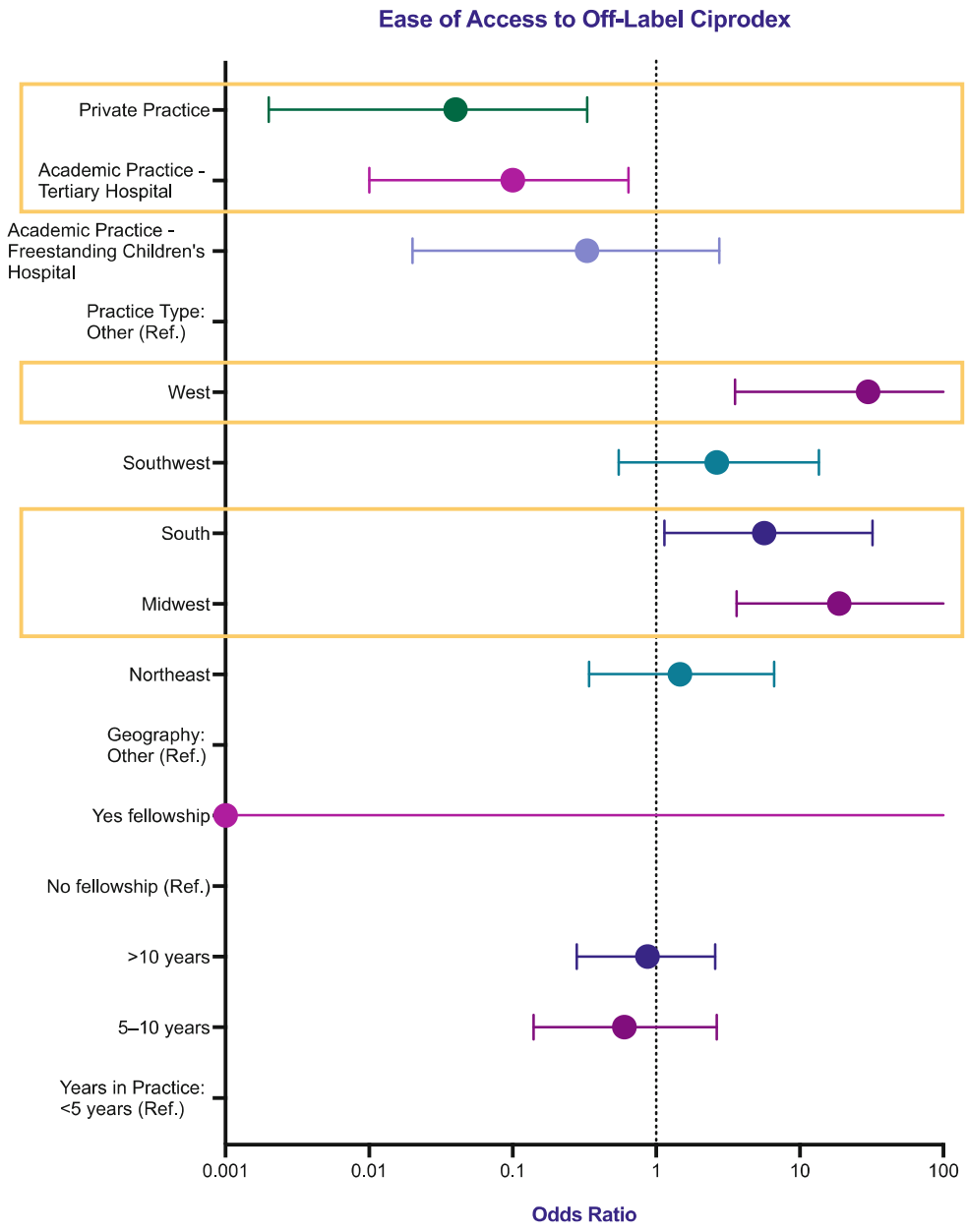

Fig. 2. Ease of access to off-label ciprofloxacin-dexamethasone across practice types. *Private practice includes respondents who selected either "Private Practice-Freestanding Children's Hospital" or "Private Practice-General Hospital." **Other includes respondents who selected either "Hospital Employed," "Non-University Teaching Hospital," "Government Hospital" or "Other." [Color figure can be viewed in the online issue, which is available at www.laryngoscope.com.]

regions and practice types. Pediatric otolaryngologists in the Midwest and West reported significantly easier access than respondents in the Northeast. We theorize that this difference may be attributed to the presence of several freestanding destination children's hospitals in the Midwest that perform a high volume of complex airway surgeries warranting more frequent off-label CPD use. The Northeast had the highest rate of responders reporting difficulty accessing off-label CPD. Providers at academic tertiary referral centers and in private practice had significantly lower likelihood of easy access to off-label CPD compared to freestanding children's hospitals. For example, one respondent, practicing at a tertiary referral center in the Northeast, reported not using CPD because "it

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is so hard to get, I wind up using something else." CPD can be expensive, which may prompt pharmacies, hospitals, and practices to limit off-label use. Administrative efforts with interdisciplinary teams may be warranted to overcome challenges in accessing CPD for patient benefit.

Patient safety is a key consideration for any off-label medication use. In our study, 85.2% of pediatric otolaryngologists ($n = 133$) felt either "Very comfortable" or "Somewhat comfortable" with the safety profile of off-label CPD. While 14.7% reported feeling "Very Uncomfortable" no adverse patient events were reported by this group, all of them use the medication at least 1–5 times per month, and all felt that the benefits outweigh the risks; although we cannot ascertain the reasoning for these respondents' discomfort, we postulate it may have been a misread given their positive responses to

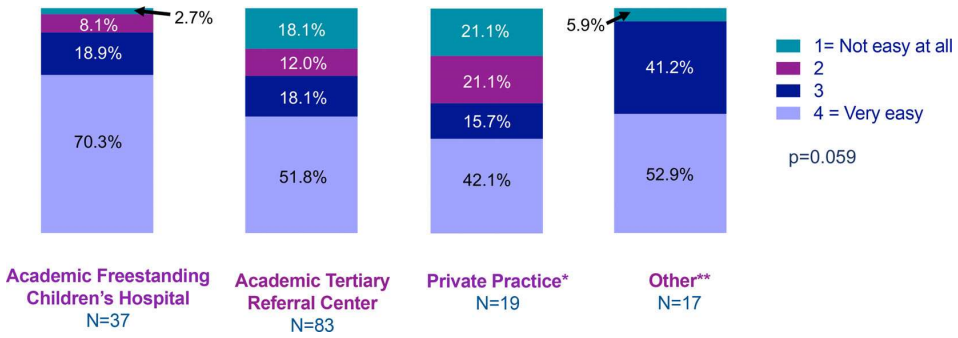

Fig. 3. Factors associated with likelihood of easy access to off-label ciprofloxacin-dexamethasone. [Color figure can be viewed in the online issue, which is available at www.laryngoscope.com.]

Fig. 4. Comfort level with off-label ciprofloxacin-dexamethasone across number of years in practice. [Color figure can be viewed in the online issue, which is available at www.laryngoscope.com.]

TABLE III.
Perceptions of and Experiences with Safety Profile of Off-Label Ciprofloxacin-Dexamethasone.

$N = 156$	All $N = 156$	<5 years $N = 31$	5–10 years $N = 28$	>10 years $N = 97$	p-Value
Benefits outweigh risks, yes (%)	155 (99.4)	31 (100.0)	28 (100.0)	96 (99.0)	0.736
Patient experience of adverse effects, yes (%)	7 (4.5)	0 (0.0)	0 (0.0)	7 (7.2)	0.077
Frequency of adverse effects ($N = 7$)					–
Once a month	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	
Once a year	1 (0.7)	0 (0.0)	0 (0.0)	1 (14.3)	
Once every few years	6 (3.8)	0 (0.0)	0 (0.0)	6 (85.7)	
Severity of adverse effects ($N = 7$)					–
1 = Not severe at all	2 (1.3)	0 (0.0)	0 (0.0)	2 (28.6)	
2	3 (1.9)	0 (0.0)	0 (0.0)	3 (42.9)	
3	2 (1.3)	0 (0.0)	0 (0.0)	2 (28.6)	
4 = Very severe	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	

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subsequent questions indicating frequent off-label use with benefits outweighing risks. Only 4.5% of respondents ($n = 7$) reported having had patients who experienced AEs. These complications seem to be mostly related to the steroid component of CPD with four responses reporting Cushingoid symptoms consistent with reports in the literature. In reviewing the safety of intranasal steroid drops and sprays in children, Ben-Dov et al. found that most AEs involved hyperglycemia and Cushing's syndrome.[15] These were more likely to occur in very young children or in those who received a prolonged course (months to years) of steroid application. Nasal drops can have some systemic absorption that may be increased in young children, especially when drops are administered lying down because the drops are not only absorbed in the respiratory tract but also swallowed.[15] Ben-Dov et al. discuss hyperglycemia as a rare, but potential complication in the literature; they describe one infant with type 1 diabetes whose blood glucose increased by 40–50 points while on intranasal CPD drops (1 drop twice daily for 1 week; FDA labeling[5] states four drops of CPD (0.14 mL) comprises 0.42 mg ciprofloxacin and 0.14 mg dexamethasone).[15] Ben-Dov et al. did not investigate tracheal or nebulized CPD. No case reports or mentions of AEs with nebulized CPD were found in the literature, and most concerns reported have arisen from nasal drops in very young infants. Further investigations to ascertain safe dosing and duration of off-label CPD administration are imperative.

One challenge in accessing off-label CPD that the authors encountered at our institution was pharmacy's concern regarding the preservative benzalkonium chloride (BAC), which has been found to cause bronchoconstriction and may have a potential risk of pneumonia and UTIs. However, in our review of the literature, a study of BAC found that the concentration required to decrease FEV_1 by 20% was 6.93 mg/mL.[17] For reference, CPD has a 0.1 mg/mL concentration of BAC, and most nebulized CPD is used as a 1:1 dilution with normal saline (yielding a 0.05 mg/mL BAC), well below this bronchoconstriction threshold. There are no case reports of AEs associated with nebulized CPD. In their cohort of 41 pediatric patients, Stephens et al. found no nebulized CPD-related complications.[8] Given the limited quantitative data available regarding the AEs of nebulized CPD use, further studies to thoroughly evaluate risk of usage are crucial. Additionally, in our survey of a large number of pediatric otolaryngologists, no respondents cited side-effect profile as a reason for not using CPD. Even those who indicated discomfort with the risk profile of CPD use it frequently and feel the benefit outweighs the risk. However, it is important to consider risks and benefits for individual patients and to minimize total dose and length of treatment, especially in young children, until further studies regarding dosing, administration, and efficacy are completed.

Our study has several limitations. Given that data were obtained through a voluntary self-reporting survey, our results may be subject to self-selective and nonresponse bias. Since not all pediatric otolaryngologists are ASPO members, our respondents may not be fully representative of the subspecialty. Our study was strengthened by the sizeable response rate of 26.6% (n = 163). However, the responses given are based on physician opinion and not objective data.

Future studies are needed to gather national-level objective data regarding the safety and clinical efficacy of off-label CPD use. For example, FEV_1 measurements before and after nebulized CPD administration may provide quantitative efficacy data. Our survey was limited in that it did not ask about AEs by route of administration. Future safety investigations regarding routes of administration, dosing, and associated complications are crucial. Efforts to standardize dosing and administration would be valuable. Additionally, future studies may assess safety and efficacy of nebulized tobramycin-dexamethasone, which may be used as an alternative to avoid the perceived risk of ciprofloxacin-resistant UTIs. Finally, off-label approval and coverage of medications often require concerted efforts from providers, pharmacists, and hospital administrators. Therefore, raising awareness about the utility, efficacy, and safety of off-label CPD is essential for continuous optimization of pediatric patient care.

CONCLUSIONS

Our survey findings (26.6% response rate) demonstrate that off-label CPD use is prevalent among pediatric otolaryngologists across the United States. While many of the pediatric otolaryngologists surveyed reported feeling comfortable with the safety of off-label CPD use, there are limited data regarding long-term safety and side effects (e.g., cushingoid symptoms, local allergic reactions) of off-label use. Given this widespread use of off-label CPD, gathering quantitative data regarding safety and clinical efficacy, and developing standardized dosing and clinical practice guidelines would be valuable.

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